

Keywords

Leisure at home / Following furniture / Adjustable tea table

Tools

Figma / Blender / STM32H743 / OV2640

Introduction

NEARBY is a followable, height-adjustable tea table for young adults who want effortless downtime. Tea, a book and a phone stay together as the user moves from an enclosed balcony to the living room. The wheeled cabinet follows, parks and adjusts its surface to the next seat.

Goal

The goal is to reduce the carrying and setup that interrupt small breaks. FOLLOW starts an indoor movement session, STAY stops the table, and HEIGHT adjusts the parked surface. The user decides where to relax; the table brings the essentials along.

Through this project,

I explore how responsive furniture can let a tea break move naturally with a person's changing place and posture.

Physical lift-box prototype · Leisure interaction concepts
Local face-matching code · Motion validation next

Background

What is a followable tea table?

A mobile, height-adjustable table keeps tea and everyday leisure items nearby. It follows the user between indoor resting places, then parks for use.

Begin with a simple break at home: tea on the balcony, reading in the living room, or relaxing in a short-stay apartment. Change the setting without carrying the table.

Project Context: Home + Short-Stay Living





HOME
Living room
+ enclosed balcony

STAY
Short-stay
accommodation

BREAK
Tea · reading
· relaxation

FOLLOW + LIFT
Move with you · fit your seat

Young Adults
Effortless downtime, wherever you sit

Main Characteristics

Move the break with you
Keep a cup, book and phone together as you switch between your indoor resting places.

Fit different ways of sitting
Use a lower surface beside the sofa or raise it for a chair. Adjust the height after parking.

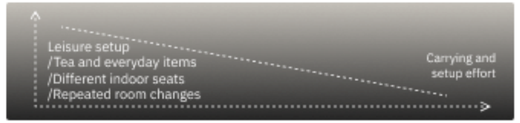
Settle in with less setup
Save a few preferred places and heights, then review the setup when staying somewhere new.

Current Situation

A relaxing break can involve small but repeated tasks: carrying drinks, moving a table and resetting a place to sit.
The project explores how furniture could move with the user.

Carrying	Moving	Resetting
/ cups + phone + book / occupied hands / repeated trips	/ table left behind / changing rooms / clearing a route	/ new sitting position / mismatched height / interrupted leisure

Opportunity Direction



Fixed furniture

The person carries items and moves the table.

Following furniture

The table follows, parks and adjusts to the seat.

Design aim: bring a complete tea break along when the user changes rooms.
Combine controllable following, a parked adjustable surface and clear status, so small moves need less manual carrying.



Basis: creator's brief - 2026-10-10. Photos: Pexels P2, P4. Cover: AI concept simulation. Research assumptions; benefits not measured.

Scenario Research

Home and short-stay leisure · Scenario study from the brief
Questions for future observation, not completed fieldwork.

Pain Point Themes

Human–Object Interaction: Carrying a complete tea break



Several items to carry

Hands already occupied



Table stays in one room

Extra trips for essentials



Drink needs a stable surface

Balancing a tray

TEA BREAK
RESET

Different seat heights

Setup starts over again

Illustrative context photographs · carrying, changing places and settling in are design hypotheses.

Human–Space Interaction: Switching the place to relax



1. Indoor movement



2. Doorway transitions



3. Leisure setup

Follow indoor routes between leisure spots.

Review each new room before using motion.

Human–Furniture Interaction: What should change?



1. Move with me



2. Fit my seat



3. Let me pause

Could following and height adjustment reduce carrying and setup?

Compare with a manual table.

Basis: creator's leisure brief. Context images: Pexels P1–P8; illustrations, not participant evidence.

User Research & Insights

User Groups

Home lounger
Young adult · tea, reading, relaxing
Switches between favourite seats

Short-stay guest
Young adult · temporary apartment
Needs a simple room setup

Shared-home user
Young adult · shared living spaces
Wants clear, controllable motion

Recruitment targets, not completed interviews.

Problem Redefinition

What interrupts an easy indoor break?
Three questions derived from the brief.

Carrying the whole setup
Can cup, book and phone move together without repeated trips?

Changing resting places
Can the table follow when comfort or light sends me elsewhere?

Fitting the next seat
Can a parked surface adapt to a sofa, chair or relaxed posture?

Design opportunity: less carrying + controllable following + a surface that fits.

Design Lens: Leisure Interruptions

- Track the work around the break: packing, carrying, repositioning and resetting.
- Compare automatic following with a tray and a manually moved side table.

POSSIBLE EXPERIENCES

Ready · Carrying · Interrupted · Relaxed

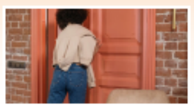
Setup tasks → Perceived effort → Break interrupted

Explore when a change of place interrupts leisure and what movement people would accept.
The relationships and feelings shown are hypotheses for a future study.

Basis: brief-led scenario analysis · validate through observation and comparative trials.

User Journey & Effort Map

BASELINE SCENARIO / Balcony tea → living-room rest with a fixed table · hypothesis

	1 PREPARE	2 CARRY	3 SETTLE	4 RELOCATE	5 RETURN	6 RESET	7 ADJUST	8 RELAX
Possible questions	*What do I need for a break?*	*Can I carry everything?*	*Where shall I sit first?*	*Shall I move to the sofa?*	*Did I leave my phone?*	*How do I move the table too?*	*Is this height comfortable?*	*Can I relax again now?*
Practical demands	Gather the tea setup	Balance several items	Choose a resting place	Change the setting	Retrieve a forgotten item	Carry and reposition	Fit the new seat	Resume the break
Emotional experience	😐	☹	😐	🙂	😐	☹	😐	🙂
Actions	 Tea preparation context	 Tray + small essentials	 Balcony break context	 Room-to-room movement	 Doorway / return route	 Next seated tea setup	 Seated reach context	 Sofa reading context

Design Concept

Solution Integration

Your tea break, on the move.
Following + height-adjustable furniture for young adults' indoor downtime.

Young adult / direct controls
FOLLOW · STAY · HEIGHT

Personal app
Save places · Set height · Check charge

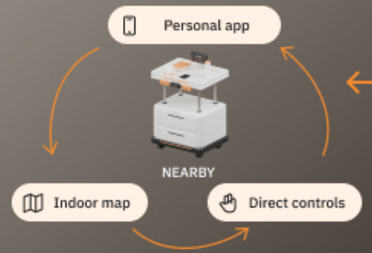
Tea table + indoor map
Tea setup · Parking places · Visible state

Design Goals

01 **Bring the whole break along.**
Keep a cup, tea flask, book and phone on one mobile surface.

02 **Move between indoor leisure spots.**
Start FOLLOW, change rooms and stop beside the next seat.

03 **Fit the surface to the moment.**
Adjust the tabletop after parking; save a preferred height.



System Integration

The tea table, direct controls and personal app share movement and height states. The map adds a few parking places for indoor breaks.

Technology

Prototype: wheeled cabinet + lift.
Next: following + parked-height tests.

Technical Research

Embedded Vision + Motion Control

On-device vision [1]
Local face matching links the device to a selected user. Identity matching and continuous person tracking are separate engineering tasks.

Four-point electric lift [2]
Four A2 motor-driven electric pushrods raise and lower the tabletop. Coordinated travel, stopping and load stability need verification.

Following + phone commands [3, 4]
The creator reports a powered base that follows a person. The proposed phone UI adds start, pause and parked-height controls.

Basic Structure / Functional Architecture



Technology Applications / Reference Examples



01 Recognise
On-device face matching
ST application [1]



02 Lift
Electric linear motion
Actuator reference [2]

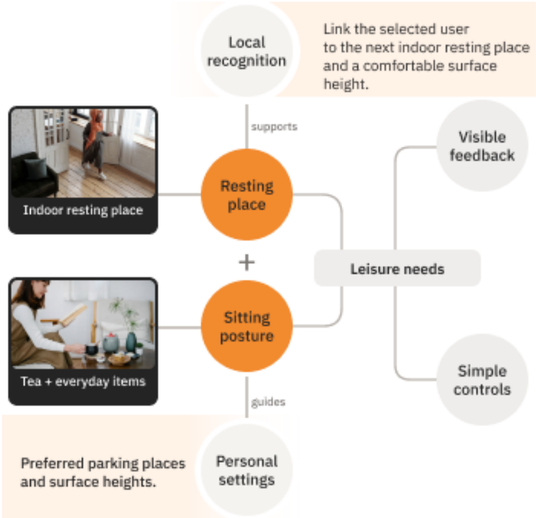


03 Move
Motor-driven wheelbase
Drive reference [3]

[1] ST / FP-AI-FACEREC1 [2] Bansbach / Electric actuators
[3] Pololu / 3pi+ 2040 [4] LINAK / Desk Control App

Concept Development

How Does NEARBY Help?



Advantages to Validate

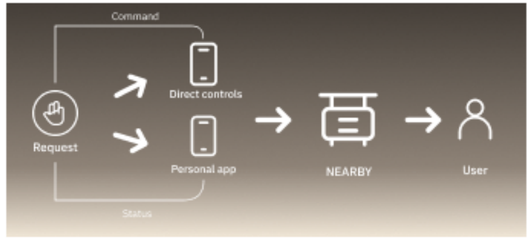
Continuity	Control	Less carrying	Comfort	Simple setup
Bring the break with you	Start and pause at any time	Keep essentials together	Fit the next sitting position	Save a place and height

Core Topic

A tea break that moves with you.
Bring leisure items from balcony to sofa, then park and fit the surface to the seat.

REQUEST
- FOLLOW starts a movement session. STAY stops.
- HEIGHT adjusts after parking.

FEEDBACK
- Following / Paused / Ready
- Visible feedback for the next action.



1 Choose FOLLOW, STAY or HEIGHT.
2 Park before reaching or changing height.
3 Pause on STAY or lost tracking; resume through a deliberate action.

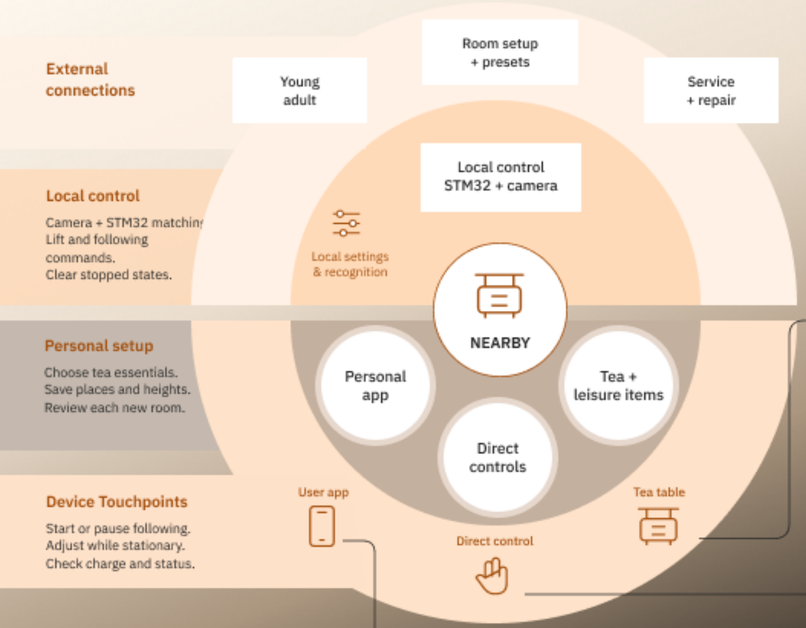
Essential Information

Next-action information
Show only what matters at this step.

- Cup · tea flask
- Book · phone
- Indoor places
- Clear routes
- Following · parked
- Height + battery

System Design

System Map



Proposed settings link a user, indoor leisure spots and preferred heights.
Following is an explicit session; reaching and lifting happen after parking.

Touchpoint Definition

Personal Touchpoint: A Complete Tea Break

Cup, book and phone travel together.
Direct controls show whether the table is following, paused or ready to use.



One tea setup
Keep cup, book and phone on the tray.

Follow on request
Start FOLLOW; use STAY to stop at a seat.

Fit the seat
Adjust only after the table is parked.

Product Functional Architecture

- Carry the leisure setup together.
- Follow between indoor resting spots.
- Adjust the surface after parking.
- Show movement, height and charge.

- FOLLOW: start a following session.
- STAY: stop; HEIGHT: set the surface.

- Review the current room setup.
- Save a place and preferred height.

Device Touchpoints

A Tea table
Mobile storage - adjustable tray
Visible movement and height state

B Direct controls
FOLLOW / STAY / HEIGHT
One-touch movement and adjustment

C Personal app
Room setup, places and presets
Review device status and charging

Essential Functions
Mobile tea storage with direct following and height controls.

Following + Lift
Carry the break between seats, then fit the surface height.

Extended Functions
Room map, saved places, height presets and charge review.

Touchpoints + Environment

Space Management

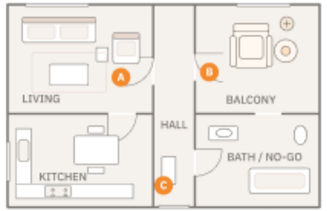
Home + short-stay review

- Identify living-room and balcony seats.
- Check doorway clearance and thresholds.
- Review the current accommodation.

Adjustable settings

- Save a few indoor parking places.
- Mark wet areas and door thresholds.
- Confirm a clear route for following.
- Pause if the target or route is unclear.

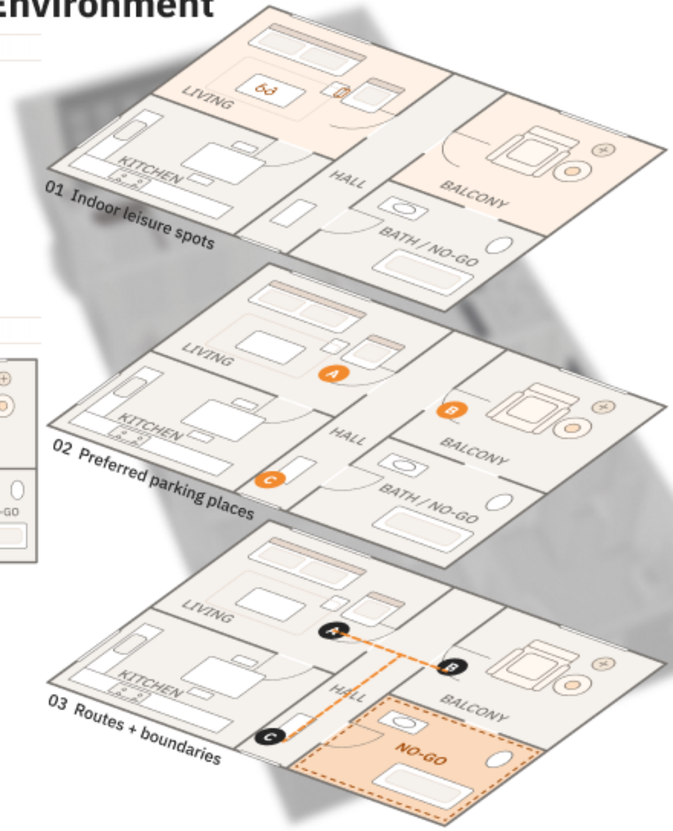
Product Layout Principle



A / Living-room sofa
Park beside the sofa with tea nearby.
Adjust the surface for a relaxed seat and keep the walking route clear.

B / Balcony chair
Use an enclosed balcony on the same level.
Review the doorway threshold before enabling a following route.

C / Charging place
Return the table to a clear charging spot.
Review the layout when moving to a new short-stay accommodation.



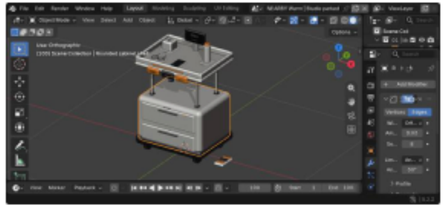
Device design

Proposed component architecture



Tea tray / four-point electric lift / following base

Prototype: wheeled cabinet and electric lift.
Next: integrated following, stopping and load tests.



Blender modelling view · concept renders

Follow
Lift
Move
Feedback

Local matching + motion control

STM32H743 + OV2640 selected for local matching.
Core code tested on PC; board testing pending.
Following is creator-reported; integrated tests next.

Visible feedback

STAY confirms a stop.
Read the status before reaching or adjusting.

Direct controls

FOLLOW / STAY / HEIGHT
Start a session, pause and adjust after parking.

Four-point lift

Four A2 pushrods lift the tray.
Adjust after parking to fit the next seat.



Tea + reading
Cup · flask · book · phone

Direct buttons
Follow · stop · height

Mobile base
Following wheel base

AI concept simulation · parked tea break; proposed controls and appearance

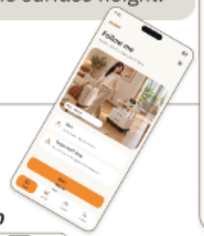
App Design

Personal leisure controls



Follow & adjust

Start following.
Park by the next seat.
Set the surface height.



Home





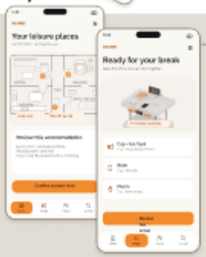
Save a tea-break preset
and recall its place and preferred height.

Room & Preset Overview

The proposed personal overview keeps indoor parking places, preferred heights and battery status together. Review the layout when staying somewhere new.

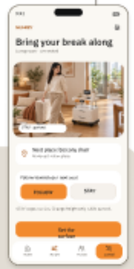
Places & tea setup

Review this room, save indoor parking places and keep tea, a book and phone ready to move.



Direct control

Start or pause, check charge and adjust a parked table.



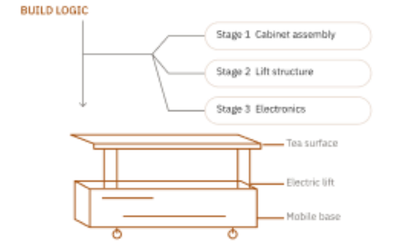
Visual Guidelines

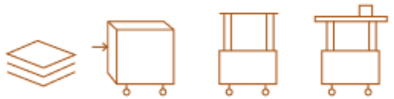
Font
IBM Plex Sans

Colour

Prototype | Following tea table

From a lift-box demonstrator to a leisure tea table: cabinet assembly, electric lift and control integration.





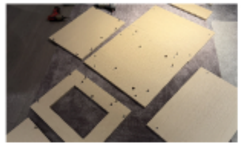
Panel kit — Cabinet → lift → integrated prototype



02–03 Assemble the cabinet and secure the drawer rails.



Mechanical assembly / steps 01–06



01 Prepare the panels



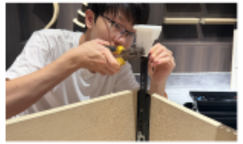
02 Fasten the cabinet



03 Fit the drawer rails



04 Mount lift hardware



05 Align the guide rails



06 Tighten the fasteners

Wiring preparation / steps 07–12



07 Solder the control board



08 Prepare actuator connections



09 Wire the control board



10 Connect the cable set



11 Route cables inside the cabinet



12 Fit the display and connectio



Original lift-box demonstrator

STM32H743 + OV2640 · fixed ROI
PC-tested core; board testing pending
Face matching supports pairing; tracking is separate.

A break that follows



Reflection + Next Test

The prototype made room-to-room leisure a concrete interaction problem. Next I would compare NEARBY with a manual side table, measuring carrying trips, setup effort, stop accuracy and drink stability. Following and lift control need testing together, with height changes only after parking.

AI concept simulation · movement behavior and drink stability not yet validated